

Analytical Validation Summary

This document summarizes the analytical method validation data for the drug product in accordance with ICH Q2(R1) Validation of Analytical Procedures. The validation covers accuracy, precision, specificity, linearity, range, and robustness as required by FDA regulations and applicable guidance documents.

Method Validation · Impurity Profile Analysis

The method is validated for the quantitative determination of impurities in the drug substance using high-performance liquid chromatography (HPLC). Validation was performed according to the acceptance criteria defined in ICH Q3A/Q3B impurities guidelines.

Acceptance Criteria (see Analytical Procedures, Table 1):

The following specification limits apply for impurities in the drug product:

- NMT 0.15% for any single impurity (see Analytical Procedures, Table 1)
- NMT 0.50% for total impurities

These limits were established based on qualification thresholds consistent with ICH Q3A guidance for daily dose. Any impurity exceeding the NMT 0.15% single-impurity specification constitutes a potential deficiency requiring immediate review.

Validation Results Summary

1. Accuracy

Recovery data demonstrated acceptable accuracy across the linear range (80-120%) with mean recovery values between 98.2% and 101.8%, meeting the acceptance criterion of 98-102% per the analytical procedure specification.

2. Precision

Repeatability (n=6) and intermediate precision (3 analysts, 3 days) were evaluated. The %RSD for repeatability was 0.8% and for intermediate precision was 1.2%, both within the NMT 2.0% acceptance criterion.

3. Linearity and Range

The analytical procedure demonstrates linearity over the range of 0.05% to 0.30% (limit of quantitation to 200% of the specification limit). The correlation coefficient (r^2) was 0.9998, meeting the acceptance criterion of not less than 0.999.

4. Stability of Analytical Solutions

Analytical solutions were found stable for at least 24 hours under ambient conditions and up to 72 hours when refrigerated, ensuring reproducibility across sample sets.

Conclusion: The analytical procedure meets all validation acceptance criteria per ICH Q2(R1) and is suitable for routine testing of the drug product impurity profile.

Cross-Reference to Analytical Procedures

For detailed impurity quantitation results, refer to Analytical Procedures document (doc_b.docx), specifically Table 1 (Impurity Profile) and Table 2 (Batch Data Summary). These tables contain the measured impurity values that are compared against the acceptance criteria stated in this Analytical Validation Summary.

Dissolution Testing

Method validation for dissolution testing was performed per USP <711>. The dissolution limit NMT 80% at 30 minutes (Q value) was evaluated in comparison to the analytical specifications document.

References

- ICH Q2(R1): Validation of Analytical Procedures: Text and Methodology
- ICH Q3A(R2): Impurities in New Drug Substances
- ICH Q3B(R2): Impurities in New Drug Products
- 21 CFR 314.50(d)(1): Specifications and analytical methods